

Video Transcript

Video 4: Embedding Terminal Window Commands in Python (03:42)

You can think of automation programs as glue that bring together a series of tasks that normally get performed independently. We'll frequently want to reach in into the shell of our operating system, as you can see here in this case, and we want to be able to do it from Python. We'll want to be able to run commands as if we were typing them directly at the command line. So, let's take a look and see how we can do that within Python. As you can see here, I have my editor open, I have a single file open called `shell.py`, and I'm going to go ahead and import a couple of packages, the operating system package, the system package, and then I'm going to define a couple of input arguments. This is something you frequently see in command line invocations of a program, and in this case we're writing them ourselves. The very first one is going to list the files, as you can see there.

The second one is going to list hidden files, a very simple thing to do simply to illustrate how to write a program that runs command line instructions. So, after that, we're going to define our two functions. The very first one is going to list files, and as you can see here, this is the command to be able to list files. Then the very next one is going to do the same thing, except that is going to list hidden files. And you can see here how we pass the command onto the shell and how we catch those results. And in this case, we're going to be printing them to the console. Then once we do that, the next step is that we're going to read those input arguments. We talked about two of them, one to list files, one to list the hidden files. And you can see here that we're going to read those arguments from the command line. The very first one would be the name of the program and the second one or the Python file, and the second one would be the argument.

Now, once we do that, you can see here that I'm going to check for what argument was passed then and if it's `list`, then I'm going to call `listfiles`. Otherwise, I'm going to request `list hidden files`, as you can see. So, I'm going to go ahead and save that. Now, I'm going to go ahead and open up the console and I'm going to go ahead and run that program. So, I'm going to go ahead and enter `python3`, the name of the program. And first, I'm not going to enter anything. As you can

see there, nothing happens. Then I'm going to go ahead and enter list for list files. And as you can see there we have a single file in that directory. And so we get a listing of the files. Now, let's go ahead and create a hidden file. So, I'm going to go ahead and enter touch and I'm just going to write here hidden. Now, if I enter list again, you'll see that I only get one file because this is the just the normal listing. But if I enter hidden, then I should be able to see both and I can. So, as you can see there, this is a pretty convenient way to be able to run commands at the command line, but to do so programmatically within our Python program. So, go ahead and give it a try, see if you can replicate, explore a little bit of what you could do, and make sure you're comfortable running command line arguments through Python.

